

Assignment 3

fahrenheit_celcius_conversion.py.

```
fahrenheit_celcius_conversion.py X
CPSC-3310 intro object-oriented > Assignment 3 > fahrenheit_celcius_conversion.py > ...
1  ABSOLUTE_ZERO_OF_CELSIUS = -273.15
2  ABSOLUTE_ZERO_OF_FAHRENHEIT = -459.67
3
4  convert_tem = float(input("Enter the temperature you want to convert: "))
5  type_tem = input("Enter the type of temperature (F for Fahrenheit, C for Celsius): ")
6  if type_tem == "F": # check the type
7      if convert_tem >= ABSOLUTE_ZERO_OF_FAHRENHEIT: # check the number
8          celsius = (convert_tem - 32) * 5 / 9
9          print(f'The temperature in Celsius is: {celsius:.2f}')
10
11     else:
12         print("Invalid temperature. Please enter a temperature above absolute zero.")
13
14 elif type_tem == "C":
15     if convert_tem >= ABSOLUTE_ZERO_OF_CELSIUS:
16         fahrenheit = (convert_tem * 9 / 5) + 32
17         print(f'The temperature in Fahrenheit is: {fahrenheit:.2f}')
18
19     else:
20         print("Invalid temperature. Please enter a temperature above absolute zero.")
21 else:
22     print("Invalid temperature type. Please enter 'F' for fahrenheit, 'C' for Celsius.")
23
```

```
fahrenheit_celcius_conversion.py X
CPSC-3310 intro object-oriented > Assignment 3 > fahrenheit_celcius_conversion.py > ...
1  ABSOLUTE_ZERO_OF_CELSIUS = -273.15
2  ABSOLUTE_ZERO_OF_FAHRENHEIT = -459.67
3
4  convert_tem = float(input("Enter the temperature you want to convert: "))
5  type_tem = input("Enter the type of temperature (F for Fahrenheit, C for Celsius): ")
6  if type_tem == "F": # check the type
7      if convert_tem >= ABSOLUTE_ZERO_OF_FAHRENHEIT: # check the number
8          celsius = (convert_tem - 32) * 5 / 9
9          print(f'The temperature in Celsius is: {celsius:.2f}')
10
11     else:
12         print("Invalid temperature. Please enter a temperature above absolute zero.")
13
14 elif type_tem == "C":
15     if convert_tem >= ABSOLUTE_ZERO_OF_CELSIUS:
16         fahrenheit = (convert_tem * 9 / 5) + 32
17         print(f'The temperature in Fahrenheit is: {fahrenheit:.2f}')
18
19     else:
20         print("Invalid temperature. Please enter a temperature above absolute zero.")
21 else:
22     print("Invalid temperature type. Please enter 'F' for fahrenheit, 'C' for Celsius.")
23
```

Problems Output Debug Console Terminal Ports

```
rs/shend/OneDrive/2026 Fall Smster/CPSC Fall/CPSC-3310 intro object-oriented/Assignment 3/fahrenheit_celcius_conversion.py"
Enter the temperature you want to convert: 100
Enter the type of temperature (F for Fahrenheit, C for Celsius): F
The temperature in Celsius is: 37.78
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented> & C:\Users\shend\AppData\Local\Programs\Python\Python313\python.exe "c:/rs/shend/OneDrive/2026 Fall Smster/CPSC Fall/CPSC-3310 intro object-oriented/Assignment 3/fahrenheit_celcius_conversion.py"
Enter the temperature you want to convert: -500.5
Enter the type of temperature (F for Fahrenheit, C for Celsius): C
Invalid temperature. Please enter a temperature above absolute zero.
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented> & C:\Users\shend\AppData\Local\Programs\Python\Python313\python.exe "c:/rs/shend/OneDrive/2026 Fall Smster/CPSC Fall/CPSC-3310 intro object-oriented/Assignment 3/fahrenheit_celcius_conversion.py"
Enter the temperature you want to convert: 50
Enter the type of temperature (F for Fahrenheit, C for Celsius): P
Invalid temperature type. Please enter 'F' for fahrenheit, 'C' for Celsius.
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented> ^C
```

student_grading.py

```
student_grading.py
C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-
oriented\Assignment 3\student_grading.py
1  ASSIGNMENT_WEIGHT = 0.3
2  QUIZ_WEIGHT = 0.2
3
4
5  exam_grade = 0
6  assignment_grade = 0
7  quiz_grade = 0
8
9  midterm_exam_grade = float(input("Please enter midterm exam grade 0 - 100: "))
10 final_exam_grade = float(input("Please enter final exam grade 0 - 100: "))
11 first_assignment_grade = float(input("Please enter first assignment grade 0 - 30: "))
12 second_assignment_grade = float(input("Please enter second assignment grade 0 - 30: "))
13 first_quiz_grade = float(input("Please enter firrt quiz grade 0 - 20: "))
14 second_quiz_grade = float(input("Please enter second quiz grade 0 - 20: "))
15
```

```
student_grading.py
CPS-3310 intro object-oriented > Assignment 3 > student_grading.py > ...
15
16 if (0 <= midterm_exam_grade <= 100) and (0 <= final_exam_grade <= 100):
17     avg_exam_grade = (midterm_exam_grade + final_exam_grade) / 2 # calculate the average value
18     exam_grade = (avg_exam_grade / 100) * EXAM_WEIGHT # convert to a percentage ratio then time the Weight value
19 else:
20     print("Invalid input. Please enter exams within the 0 - 100 range.")
21 if (0 <= first_assignment_grade <= 30) and (0 <= second_assignment_grade <= 30):
22     avg_assignment_grade = (first_assignment_grade + second_assignment_grade) / 2
23     assignment_grade = (avg_assignment_grade / 30) * ASSIGNMENT_WEIGHT
24 else:
25     print("Invalid input. Please enter assignments within the 0 - 30 range.")
26 if (0 <= first_quiz_grade <= 20) and (0 <= second_quiz_grade <= 20):
27     avg_quiz_grade = (first_quiz_grade + second_quiz_grade) / 2
28     quiz_grade = (avg_quiz_grade / 20) * QUIZ_WEIGHT
29 else:
30     print("Invalid input. Please enter quizzes within the 0 - 20 range.")
31
32 # add the final ratio then back to the percentage ratio
33 final_grade_ratio = (exam_grade + assignment_grade + quiz_grade)
34 print(f'The students final grade is {final_grade_ratio:.2%}')
35
```

```
student_grading.py X
CPSC-3310 intro object-oriented > Assignment 3 > student_grading.py > ...
15
16 ∨ if (0 <= midterm_exam_grade <= 100) and (0 <= final_exam_grade <= 100):
17     avg_exam_grade = (midterm_exam_grade + final_exam_grade) / 2 # calculate the average value
18     exam_grade = (avg_exam_grade / 100) * EXAM_WEIGHT # convert to a percentage ratio then time the Weight value
19 ∨ else:
20     print("Invalid input. Please enter exams within the 0 - 100 range.")
21 ∨ if (0 <= first_assignment_grade <= 30) and (0 <= second_assignment_grade <= 30):

Problems Output Debug Console Terminal Ports
rs/shend/OneDrive/2026 Fall Smster/CPSC Fall/CPSC-3310 intro object-oriented/Assignment 3/student_grading.py"
Please enter midterm exam grade 0 - 100: 85
Please enter final exam grade 0 - 100: 92
Please enter first assignment grade 0 - 30: 25
Please enter second assignment grade 0 - 30: 28
Please enter firrt quiz grade 0 - 20: 19
Please enter second quiz grade 0 - 20: 17
The students final grade is 88.75%
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented> & C:\Users\shend\AppData\Local\Programs\Python\
rs/shend/OneDrive/2026 Fall Smster/CPSC Fall/CPSC-3310 intro object-oriented/Assignment 3/student_grading.py"
Please enter midterm exam grade 0 - 100: 110
Please enter final exam grade 0 - 100: 90
Please enter first assignment grade 0 - 30: 40
Please enter second assignment grade 0 - 30: 20
Please enter firrt quiz grade 0 - 20: -10
Please enter second quiz grade 0 - 20: 19
Invalid input. Please enter exams within the 0 - 100 range.
Invalid input. Please enter assignments within the 0 - 30 range.
Invalid input. Please enter quizzes within the 0 - 20 range.
The students final grade is 0.00%
PS C:\Users\shend\OneDrive\2026 Fall Smster\CPSC Fall\CPSC-3310 intro object-oriented> |
```

Turtle hit-target game.py

```
the Turtle hit-target game.py X
CPSC-3310 intro object-oriented > Assignment 3 > the Turtle hit-target game.py > ...
1 # Hit the Target Game
2 import turtle
3
4 # Named constants
5 SCREEN_WIDTH = 600 # Screen width
6 SCREEN_HEIGHT = 600 # Screen height
7 TARGET_LLEFT_X = 100 # Target's lower-left X
8 TARGET_LLEFT_Y = 250 # Target's lower-left Y
9 TARGET_WIDTH = 25 # Width of the target
10 FORCE_FACTOR = 60 # Arbitrary force factor
11 PROJECTILE_SPEED = 1 # Projectile's animation speed
12 NORTH = 90 # Angle of north direction
13 SOUTH = 270 # Angle of south direction
14 EAST = 0 # Angle of east direction
15 WEST = 180 # Angle of west direction
16
17 # Setup the window.
18 turtle.setup(SCREEN_WIDTH, SCREEN_HEIGHT)
19
```

the Turtle hit-target game.py X

CPSC-3310 intro object-oriented > Assignment 3 > the Turtle hit-target game.py > ...

```
19
20 # Draw the target.
21 turtle.hideturtle()
22 turtle.speed(0)
23 turtle.penup()
24 turtle.goto(TARGET_LLEFT_X, TARGET_LLEFT_Y)
25 turtle.pendown()
26 turtle.setheading(EAST)
27 turtle.forward(TARGET_WIDTH)
28 turtle.setheading(NORTH)
29 turtle.forward(TARGET_WIDTH)
30 turtle.setheading(WEST)
31 turtle.forward(TARGET_WIDTH)
32 turtle.setheading(SOUTH)
33 turtle.forward(TARGET_WIDTH)
34 turtle.penup()
35
36 # Center the turtle.
37 turtle.goto(0, 0)
38 turtle.setheading(EAST)
39 turtle.showturtle()
40 turtle.speed(PROJECTILE_SPEED)
41
42 # Get the angle and force from the user.
43 angle = turtle.numinput("Input", "Enter the projectile's angle",
44 | | | minval=0, maxval=360)
45 force = turtle.numinput("Input", "Enter the launch force (1-10)",
46 | | | minval=1, maxval=10)
47
```

the Turtle hit-target game.py X

CPSC-3310 intro object-oriented > Assignment 3 > the Turtle hit-target game.py > ...

```
36 # Center the turtle.
37 turtle.goto(0, 0)
38 turtle.setheading(EAST)
39 turtle.showturtle()
40 turtle.speed(PROJECTILE_SPEED)
41
42 # Get the angle and force from the user.
43 angle = turtle.numinput("Input", "Enter the projectile's angle",
44 | | | minval=0, maxval=360)
45 force = turtle.numinput("Input", "Enter the launch force (1-10)",
46 | | | minval=1, maxval=10)
47
48 # Calculate the distance.
49 distance = force * FORCE_FACTOR
50
51 # Set the heading.
52 turtle.setheading(angle)
53
54 # Launch the projectile.
55 turtle.pendown()
56 turtle.forward(distance)
57
58 # Did it hit the target?
59 if (turtle.xcor() >= TARGET_LLEFT_X and
60 | turtle.xcor() <= (TARGET_LLEFT_X + TARGET_WIDTH) and
61 | turtle.ycor() >= TARGET_LLEFT_Y and
62 | turtle.ycor() <= (TARGET_LLEFT_Y + TARGET_WIDTH)):
63     print('Target hit!')
```

```

# Did it hit the target?
if (turtle.xcor() >= TARGET_LLEFT_X and
    turtle.xcor() <= (TARGET_LLEFT_X + TARGET_WIDTH) and
    turtle.ycor() >= TARGET_LLEFT_Y and
    turtle.ycor() <= (TARGET_LLEFT_Y + TARGET_WIDTH)):
    print('Target hit!')
else:
    print('You missed the target.')
    if turtle.xcor() > (TARGET_LLEFT_X + TARGET_WIDTH):
        print("Try a grater angle")
    if turtle.xcor() < (TARGET_LLEFT_X ):
        print("Try a lesser angle")
    if turtle.ycor() > (TARGET_LLEFT_Y + TARGET_WIDTH):
        print("Try lesser force")
    if turtle.ycor() < (TARGET_LLEFT_Y):
        print("Try more force")

turtle.done()

```

